evaluate_jsdm.R

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`evaluate_jsdm` *Evaluate JSDM Model Performance*

Description

Evaluates a fitted sjSDM model and returns comprehensive performance metrics at both model and species level.

Usage

```
evaluate_jsdm(mod_jsdm = NULL)
```

Arguments

`mod_jsdm`	A fitted sjSDM model object. Must be of class 'sjSDM'.

Details

For binomial models, species-level classification metrics (AUC, Accuracy, LogLoss) are computed using a 0.5 probability threshold for binary predictions. For other model families, RMSE is computed per species.

Convergence is assessed via [check_convergence_jsdm()], which analyses the training loss history. A 'linear_trend_slope' < 0.01 and 'median_diff' < 1 in the returned 'convergence' element indicate that the model has converged.

Value

A list with three elements: - 'model': Named numeric vector of R-squared values (McFadden, Nagelkerke) - 'species': A tibble with one row per species and columns: species, AUC, Accuracy, LogLoss (binomial) or RMSE (other families) - 'convergence': A list from [check_convergence_jsdm()] with 'linear_trend_slope', 'median_diff', 'convergence_plot', and 'note'

See Also

sjSDM::Rsquared, Metrics::auc, check_convergence_jsdm

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